

WEEK 7 REPORT

Student Performance – Data Cleaning & Transformation

-done by **Sai Meera R** from **B.sc Data Science & AI**

Introduction

The objective of Week 7 was to clean, transform, and prepare the Student Performance dataset for analysis in Power BI. The dataset was inspected for data quality issues, necessary calculated columns were created, subject data was transformed using unpivoting, and DAX measures were created for performance analysis.

The final dataset was validated to ensure that the calculated values, classifications, relationships, and measures were working correctly.

Dataset Import and Inspection

The Student Performance dataset was imported into Power BI and opened in Power Query Editor for data preparation.

The dataset was inspected to understand the available numerical and categorical attributes. Data types, missing values, errors, and duplicate records were also reviewed as part of the initial data-quality check.

Student Performance dataset in Power Query Editor

StudentsPerformance (Week 7 Dataset)

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75
female	group B	associate's degree	standard	none	71	83	78
female	group B	some college	standard	completed	88	95	92
male	group B	some college	free/reduced	none	40	43	39
male	group D	high school	free/reduced	completed	64	64	67
female	group B	high school	free/reduced	none	38	60	50
male	group C	associate's degree	standard	none	58	54	52
male	group D	associate's degree	standard	none	40	52	43
female	group B	high school	standard	none	65	81	73
male	group A	some college	standard	completed	78	72	70
female	group A	master's degree	standard	none	50	53	58
female	group C	some high school	standard	none	69	75	78
male	group C	high school	standard	none	88	89	86
female	group B	some high school	free/reduced	none	18	32	28
male	group C	master's degree	free/reduced	completed	46	42	46
female	group C	associate's degree	free/reduced	none	54	58	61

The data in the preview has been truncated due to size limits.

Extract Table Using Examples

Load

Transform Data

Cancel

Data Cleaning and Transformation

The dataset was cleaned and prepared for further analysis. The relevant columns were reviewed, data types were checked, and column names were maintained clearly for analysis.

Categorical attributes such as Gender, Race/Ethnicity, Parental Level of Education, Lunch, and Test Preparation Course were reviewed for consistency.

Cleaned Student Performance dataset

= Table.TransformColumns(#"Renamed Columns",{{"Parental level of Education", Text.Proper, type text}})									
	A ⁰ _C Race/Ethnicity	A ⁰ _C Parental level of Education	A ⁰ _C Lunch	A ⁰ _C Test preparation course	I ² ₃ Math Score	I ² ₃ Reading score			
	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	
1	group B	Bachelor'S Degree	standard	none		72		7	
2	group C	Some College	standard	completed		69		9	
3	group B	Master'S Degree	standard	none		90		9	
4	group A	Associate'S Degree	free/reduced	none		47		5	
5	group C	Some College	standard	none		76		7	
6	group B	Associate'S Degree	standard	none		71		8	
7	group B	Some College	standard	completed		88		9	
8	group B	Some College	free/reduced	none		40		4	
9	group D	High School	free/reduced	completed		64		6	
10	group B	High School	free/reduced	none		38		6	
11	group C	Associate'S Degree	standard	none		58		5	
12	group D	Associate'S Degree	standard	none		40		5	
13	group B	High School	standard	none		65		8	
14	group A	Some College	standard	completed		78		7	
15	group A	Master'S Degree	standard	none		50		5	
16	group C	Some High School	standard	none		69		7	
17	group C	High School	standard	none		88		8	
18	group B	Some High School	free/reduced	none		18		3	
19	group C	Master'S Degree	free/reduced	completed		46		4	
20	group C	Associate'S Degree	free/reduced	none		54		5	
21	group D	High School	standard	none		66		6	
22	group B	Some College	free/reduced	completed		65		7	
23	group D	Some College	standard	none		44		5	
24	group C	Some High School	standard	none		69		7	
25	group D	Some High School	free/reduced	completed		74		7	

Creation of Total Score Column

A Total Score column was created by adding the Math Score, Reading Score, and Writing Score for each student.

Formula:

Total Score = Math Score + Reading Score + Writing Score

Creation of Total Score column

Custom Column

Add a column that is computed from the other columns.

New column name

Total

Custom column formula ①

= [Math Score]+[Reading score]+[Writing score]

Available columns

Student ID
Gender
Race/Ethnicity
Parental level of Education
Lunch
Test preparation course
Math Score

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Creation of Average Score Column

An Average Score column was created to calculate the average performance of each student across the three subjects.

Formula:

Average Score = Total Score / 3

Creation of Average Score column

Custom Column

Add a column that is computed from the other columns.

New column name

Average

Custom column formula ①

= [Total]/3

Available columns

Student ID
Gender
Race/Ethnicity
Parental level of Education
Lunch
Test preparation course
Math Score

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

Performance Grade and Pass/Fail Classification

A Performance Grade column was created using conditional logic to classify student performance into different levels:

- **A** – High performance
- **B** – Good performance
- **C** – Average performance
- **F** – Low performance

A Pass/Fail classification was also created using the selected passing criterion. This allows students to be grouped according to their academic performance.

Performance Grade Classification

A **Performance Grade** column was created using conditional logic to classify students based on their overall performance.

- **A** – High performance
- **B** – Good performance
- **C** – Average performance
- **F** – Low performance

Performance Grade classification

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Performance Grade

	Column Name	Operator	Value		Output
If	Average	is greater than or...	90	Then	Grade A
Else If	Average	is greater than or...	75	Then	Grade B
Else If	Average	is greater than or...	50	Then	Grade C
Else If	Average	is greater than or...	40	Then	Grade D
Else If	Average	is less than	40	Then	Grade F

Add Clause

Else

ABC 123

null

OK

Cancel

Pass/Fail Classification

A Pass/Fail column was created using the selected passing criterion. Students were classified into two categories:

- Pass
- Fail

This classification helps identify students who meet the defined performance criterion.

Pass/Fail classification

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Pass/Fail

	Column Name	Operator	Value		Output
If	Average	is greater than	40	Then	Pass

Add Clause

Else

Fail

OK

Cancel

Data Preparation Using Duplicate Queries

To organize the student performance data for analysis, the original dataset was duplicated into separate queries.

Student Marks Table

A duplicate of the original dataset was created and named **Student Marks**. This table was used specifically for subject-wise performance analysis.

The Math Score, Reading Score, and Writing Score columns were selected and unpivoted. The resulting columns were organized as:

- Student ID
- Subject
- Score

This structure allows student performance to be analyzed separately for each subject.

Student Marks table after unpivoting subject scores

The screenshot shows a data table with the following structure:

Student ID	Attribute	Value
1	Math Score	72
2	Reading score	72
3	Writing score	74
4	Math Score	69
5	Reading score	90
6	Writing score	88
7	Math Score	90
8	Reading score	95
9	Writing score	93
10	Math Score	47
11	Reading score	57
12	Writing score	44
13	Math Score	76
14	Reading score	78
15	Writing score	75
16	Math Score	71
17	Reading score	83
18	Writing score	78
19	Math Score	88
20	Reading score	95
21	Writing score	92
22	Math Score	40
23	Reading score	43
24	Writing score	39
25	Math Score	64

The interface includes a 'Query Settings' panel on the right with 'Properties' and 'Applied Steps' sections. The 'Applied Steps' list includes 'Unpivoted Columns'.

Student Details Table

Another duplicate of the original dataset was created and named **Student Details**.

The Math Score, Reading score, and Writing score columns were removed from this table so that it could focus on the students' demographic and other descriptive information without repeating the subject-score fields.

Student Details table after removing subject score column

The screenshot shows a data table with the following structure:

Student ID	Gender	Race/Ethnicity	Parental level of Education	Lunch	Test preparation course
1	female	group B	Bachelor's Degree	standard	none
2	female	group C	Some College	standard	completed
3	female	group B	Master's Degree	standard	none
4	male	group A	Associate's Degree	free/reduced	none
5	male	group C	Some College	standard	none
6	female	group B	Associate's Degree	standard	none
7	female	group B	Some College	standard	completed
8	male	group B	Some College	free/reduced	none
9	male	group D	High School	free/reduced	completed
10	female	group B	High School	free/reduced	none
11	male	group C	Associate's Degree	standard	none
12	male	group D	Associate's Degree	standard	none
13	female	group B	High School	standard	none
14	male	group A	Some College	standard	completed
15	female	group A	Master's Degree	standard	none
16	female	group C	Some High School	standard	none
17	male	group C	High School	standard	none
18	female	group B	Some High School	free/reduced	none
19	male	group C	Master's Degree	free/reduced	completed
20	female	group C	Associate's Degree	free/reduced	none
21	male	group D	High School	standard	none
22	female	group B	Some College	free/reduced	completed
23	male	group D	Some College	standard	none
24	female	group C	Some High School	standard	none
25	male	group C	Bachelor's Degree	free/reduced	completed

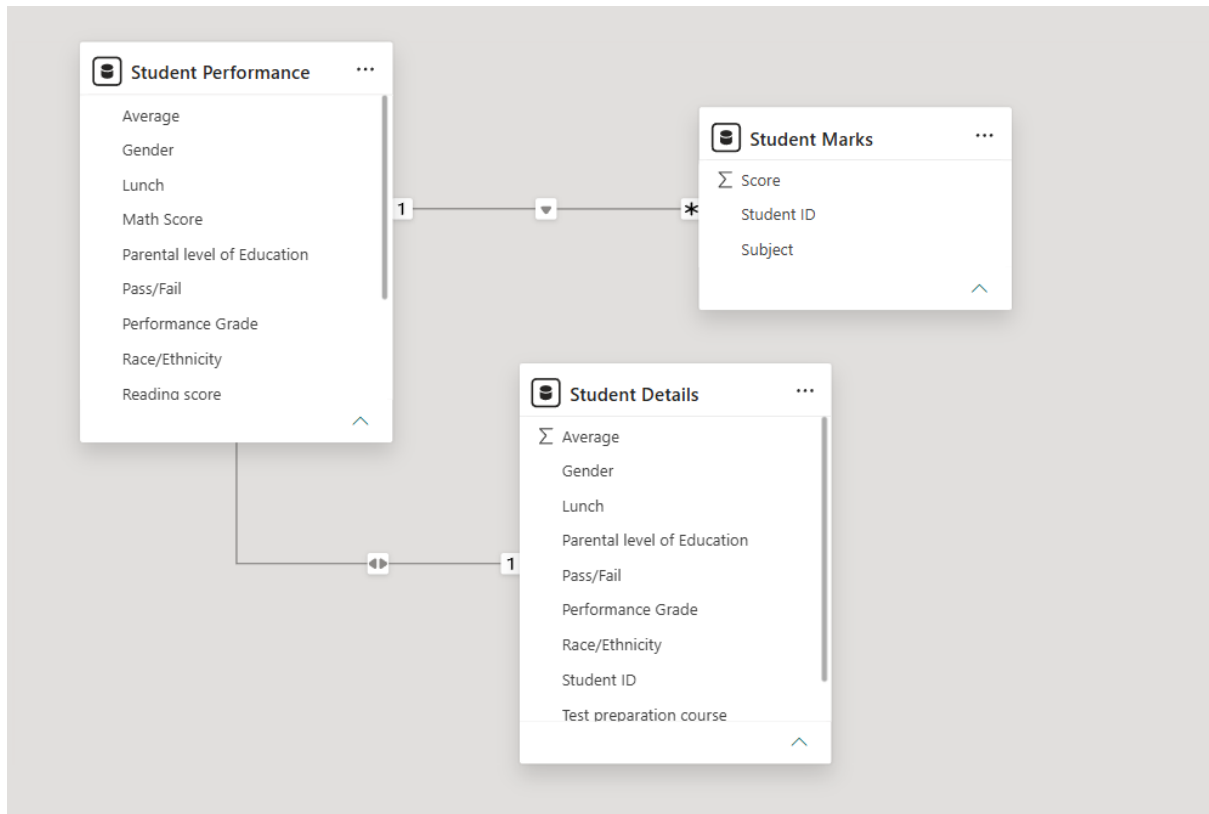
The interface includes a 'Query Settings' panel on the right with 'Properties' and 'Applied Steps' sections. The 'Applied Steps' list includes 'Removed Columns1'.

Data Model Preparation

The transformed data was loaded into the Power BI data model. The relevant tables and relationships were reviewed to ensure that the data could be used correctly for analysis.

Unnecessary data was avoided, and the transformed subject-level data was retained for further analysis.

Student Performance data model



DAX Measures

Five DAX measures were created in the Measure table to support performance analysis:

1. Average Math Score
2. Average Reading Score
3. Average Writing Score
4. Average Score
5. Overall Pass %

These measures provide dynamic calculations that can later be used in Power BI visuals and dashboard analysis.

Average Score

Average Score =

AVERAGE('Student Performance'[Total])

This measure calculates the average total score of the students.

Average Math Score

Average Math Score =

AVERAGE('Student Performance'[Math Score])

This measure calculates the average score obtained in Mathematics.

Average Reading Score

Average Reading Score =

AVERAGE('Student Performance'[Reading score])

This measure calculates the average score obtained in Reading.

Average Writing Score

Average Writing Score =

AVERAGE('Student Performance'[Writing score])

This measure calculates the average score obtained in Writing.

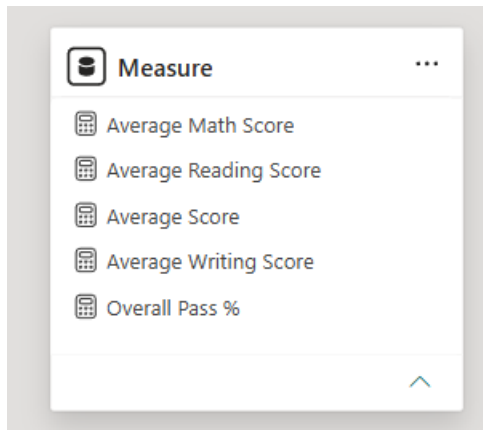
Overall Pass %

Overall Pass % =

DIVIDE(
 CALCULATE(
 COUNTROWS('Student Performance'),
 'Student Performance'[Pass/Fail] = "Pass"
),
 COUNTROWS('Student Performance'),
 0
)

This measure calculates the percentage of students who passed according to the defined pass criterion.

DAX measures created for Student Performance analysis



Validation

After completing the transformations and DAX measures, the results were validated.

The calculated Total and Average values were checked, Pass/Fail and Performance Grade classifications were verified, the unpivoted subject data was reviewed, and the DAX measures were checked to ensure that they returned valid results.

Conclusion

Week 7 focused on preparing the Student Performance dataset for analysis using Power BI. The dataset was cleaned and transformed, calculated attributes such as Total, Average, Performance Grade, and Pass/Fail were created, and subject-level data was reshaped using unpivoting.

Five DAX measures were also created to calculate important academic performance indicators. After validation, the dataset and data model were confirmed to be ready for dashboard development and analysis in Week 8.

